Factoriza los siguientes polinomios:

$$a)\ x^4 + 3x^3 - 3x^2 - 11x - 6 =$$

	1	+3	-3	-11	-6
-1		-1	-2	+5	+6
	1	+2	-5	-6	0
-1		-1	-1	+6	
	1	+1	-6	0	
2		+2	+6		
	1	+3	0		

$$Solución: x^4 + 3x^3 - 3x^2 - 11x - 6 = (x+1)^2 \cdot (x-2) \cdot (x+3)$$

$$b)\ x^4 + 6x^3 + 9x^2 - 4x - 12 =$$

	1	+6	+9	-4	-12
-3		-3	-9	0	+12
	1	+3	0	-4	0
-2		-2	-2	+4	
	1	+1	-2	0	
-2		-2	+2		
	1	-1	0		

$$Solución: x^4 + 6x^3 + 9x^2 - 4x - 12 = (x+3) \cdot (x+2)^2 \cdot (x-1)$$

$$c)\ x^4 + x^3 - 19x^2 - 49x - 30 =$$

	1	+1	-19	-49	-30
-3		-3	+6	+39	+30
	1	-2	-13	-10	0
-2		-2	+8	+10	
	1	-4	-5	0	
-1		-1	+5		
	1	-5	0		

$$Solución: x^4 + x^3 - 19x^2 - 49x - 30 = (x+3)(x+2)(x+1)(x-5)$$

$$d)\ x^4 + 10x^3 + 37x^2 + 60x + 36 =$$

	1	+10	+37	+60	+36
-3		-3	-21	-48	-36
	1	+7	+16	+12	0
-3		-3	-12	-12	
	1	+4	+4	0	
-2		-2	-4		
	1	+2	0		

$$Solución: x^4 + 10x^3 + 37x^2 + 60x + 36 = (x+3)^2 \cdot (x+2)^2$$

$$e)\ x^4 - 2x^2 + 1 =$$

	1	0	-2	0	+1
-1		-1	+1	+1	-1
	1	-1	-1	+1	0
-1		-1	+2	-1	
	1	-2	+1	0	
1		1	-1		
	1	-1	0		

$$Solución: x^4 - 2x^2 + 1 = (x+1)^2 \cdot (x-1)^2$$

$$f)\ x^4 + 3x^3 - 5x^2 - 3x + 4 =$$

	1	+3	-5	-3	+4
-4		-4	+4	+4	-4
	1	-1	-1	+1	0
-1		-1	+2	-1	
	1	-2	+1	0	
1		1	-1		
	1	-1	0		

$$Solución: x^4 + 3x^3 - 5x^2 - 3x + 4 = (x+4) \cdot (x+1) \cdot (x-1)^2$$

$$g)\ x^4 + 2x^3 - 12x^2 + 14x - 5 =$$

	1	+2	-12	+14	-5
-5		-5	+15	-15	+5
	1	-3	+3	-1	0
1		1	-2	+1	
	1	-2	+1	0	
1		1	-1		
	1	-1	0		

$$Solución: x^4 + 2x^3 - 12x^2 + 14x - 5 = (x+5) \cdot (x-1)^3$$

$$h)\ x^5 + 6x^4 + 5x^3 - 24x^2 - 36x =$$

Sacamos factor común:

$$x^5 + 6x^4 + 5x^3 - 24x^2 - 36x = x(x^4 + 6x^3 + 5x^2 - 24x - 36)$$

	1	+6	+5	-24	-36
-3		-3	-9	+12	+36
	1	+3	-4	-12	0
-3		-3	0	+12	
	1	0	-4	0	
-2		-2	+4		
	1	-2	0		

$$Solución: x^5 + 6x^4 + 5x^3 - 24x^2 - 36x = x(x+3)^2 \cdot (x+2) \cdot (x-2)$$

$$i)\ x^5 + 2x^4 - 3x^3 - 8x^2 - 4x =$$

Sacamos factor común:

$$x^5 + 2x^4 - 3x^3 - 8x^2 - 4x = x(x^4 + 2x^3 - 3x^2 - 8x - 4)$$

	1	+2	-3	-8	-4
-2		-2	0	+6	+4
	1	0	-3	-2	0
-1		-1	+1	+2	
	1	-1	-2	0	
-1		-1	+2		
	1	-2	0		

$$Solución: x^5 + 2x^4 - 3x^3 - 8x^2 - 4x = x(x+2) \cdot (x+1)^2 \cdot (x-2)$$

$$j)\ x^6 + 2x^5 - 3x^4 - 4x^3 + 4x^2 =$$

Sacamos factor común:

$$x^6 + 2x^5 - 3x^4 - 4x^3 + 4x^2 = x^2 \cdot (x^4 + 2x^3 - 3x^2 - 4x + 4)$$

	1	+2	-3	-4	+4
-2		-2	0	+6	-4
	1	0	-3	+2	0
-2		-2	+4	-2	
	1	-2	+1	0	
1		1	-1		
	1	-1	0		

$$Solución: x^6 + 2x^5 - 3x^4 - 4x^3 + 4x^2 = x^2 \cdot (x+2)^2 \cdot (x-1)^2$$